

Smart Assistive Glasses for the Visually Impaired

Group Members: Jinlin Li, Yizhe Shen, Haobo Fang,
Zhuoyang Chen, Guanyu Xu, Ruopu Dong

I. Executive Summary

Globally, more than 250 million people live with vision impairments, and their most prominent challenges are mobility and safety in daily navigation[1]. In everyday environments, people with vision impairments encounter frequent barriers, on average about 12 obstacles per kilometer, such as poles, street furniture, and parked cars[2]. These conditions force slower walking speeds, reduced independence, and greater risks of accidents and falls, which in turn lower confidence and discourage outdoor activity[2]. Our project proposes **Smart Assistive Glasses for the Visually Impaired** to address this challenge. The system will provide **real-time obstacle-aware navigation and scene description (Figure 1)**, enabling safer, more confident movement in both familiar and unfamiliar environments. The system will also support voice wake-up, audio guidance, and vibromotor obstacle indicator, making it easy for users to interact hands-free. Key considerations include power management, privacy, and user safety, with optional features such as fall alerts to enhance reliability in daily use. The importance of this solution is its ability to enhance independence, reduce reliance on costly or impractical aids, and improve quality of life by making navigation safer, more efficient, and less stressful. The project objective is clear: create a practical, user-centered assistive device that integrates embedded intelligence into a lightweight, wearable form factor. Our approach combines sensing technologies with local AI processing to deliver timely feedback without reliance on connectivity to the cloud. The result will be a functional prototype and evaluation demonstrating that such a system can be implemented in an **efficient, practical, and safe way**.

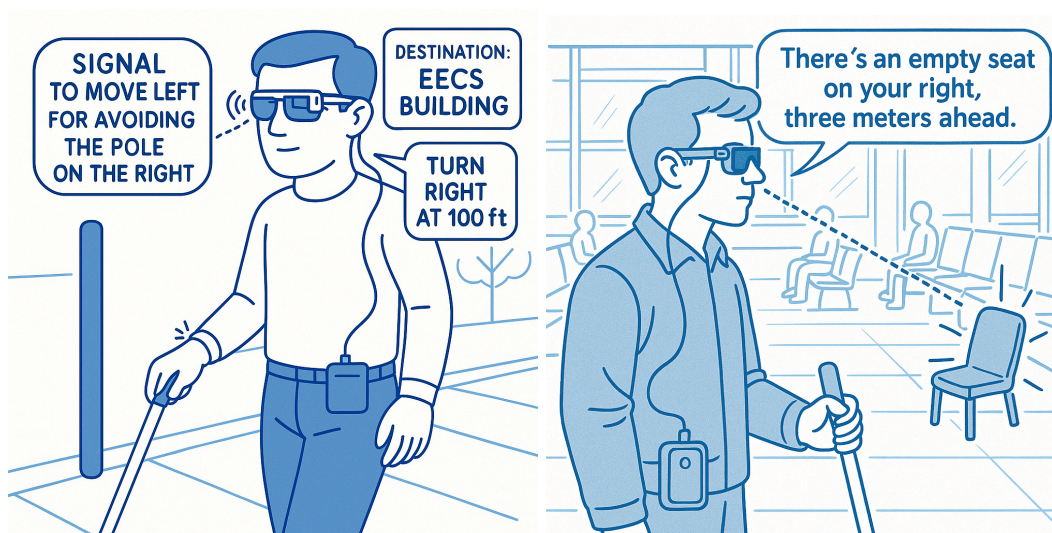


Figure 1: Scene illustration of Obstacle-aware navigation (left) and Scene description (right)

- [1]: <https://www.science.org/doi/10.1126/scirobotics.abg6594>
- [2]: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10572964/>
- [3]: <https://www.nature.com/articles/s41467-025-58085-x>
- [4]: <https://www.igdf.org.uk/about-us/facts-and-figures>

II. High-Level Description

A. Design Intent

Mobility and safety are persistent challenges for people with visual impairments. Navigating everyday environments often involves avoiding static and moving obstacles, crossing streets, and boarding buses. These tasks are especially difficult when infrastructure lacks tactile paving, audible signals, or warning strips. The need for a reliable, practical mobility solution is particularly significant for visually impaired individuals.

We intend to design a system that reduces the burden of unsafe or uncertain travel and builds confidence for individuals with visual impairments to travel independently by providing accurate, real-time guidance with minimal effort from the user. By enabling **safe obstacle detection, effective navigation, and real-time accessible interaction** in both familiar and unfamiliar environments, the system will improve confidence, enhance independence, and increase quality of life. To be successful, the system must be lightweight, affordable, power-efficient, and user-friendly while providing robust performance in diverse real-world conditions. Reliability requires accurate detection of both static and dynamic obstacles, while privacy and safety demand **responsive and local data processing** rather than dependence on cloud connectivity. The ultimate design goal is a practical and easy-to-use product that integrates seamlessly into daily mobility and empowers users to move through the world more safely and independently.

B. Proposed Solution

Solution 1: Traditional Aids

Traditional aids like white canes and guide dogs remain the most common mobility tools, but they only offer partial solutions. Canes provide ground-level feedback but cannot detect fast-moving or elevated obstacles, and collision rates remain high in crowded or dynamic environments[3]. They also cannot help users identify specific objects or contextual details in unfamiliar areas. Guide dogs provide more assistance but are extremely scarce worldwide and costly ($\approx$ \$42,000 to train), with limited working lifespans[4]. While valuable, these aids fall short of delivering scalable and comprehensive mobility support.

Solution 2: Smart Assistive Glasses for the Visually Impaired (Our solution)

Our project proposes Smart Assistive Glasses for the Visually Impaired, specifically designed to overcome these gaps. In this design, the system is divided into two main components: a lightweight **smart glasses** and a portable embedded **base station** (Figure 2).

The glasses module integrates a speaker and microphone to provide real-time audio guidance and natural interaction through voice commands. An onboard MCU continuously monitors audio for wake word detection using a hardware-optimized, low-power processing pipeline. To address privacy concerns, a small LED indicator signals when the onboard wide-angle camera is operating. The camera provides continuous video input for obstacle detection. The vibromotor provides instant obstacle indication.

The base station acts as the computational hub. Built on a Jetson embedded platform, it executes local AI models for **natural language processing**, **real-time obstacle detection**, and **scene understanding**, providing timely feedback for safe navigation and scene description. A GPS receiver gets positioning data, while an open-source navigation software will guide the user to the desired location. The base station also contains power management and maintains a wired connection to the glasses, ensuring stable data transfer, reduced latency, and longer operating times compared to wireless alternatives.

By combining lightweight wearable sensing with powerful embedded computation, this design overcomes the trade-offs of existing aids. It avoids the limited perception of canes, the cost and scarcity of guide dogs, and, compared with commercial AI glasses, our smart assistive glass provides privacy by limiting all computations locally. The result will be a practical, user-centered prototype that enhances mobility, builds user confidence, and supports independent navigation in both familiar and unfamiliar settings.

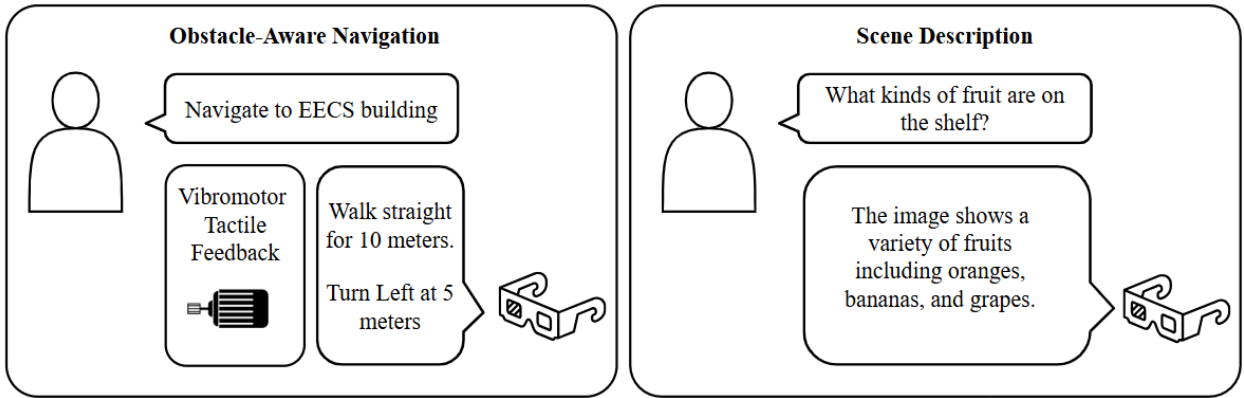


Figure 2: System Architecture of the Smart Assistive Glasses for the Visually Impaired

III. Implementation Details

A. Design Overview

The proposed system begins operation with **wake word detection** on the smart glasses, implemented using an open-source [WakeNet](#) model optimized for the ESP32-S3 microcontroller. Upon detecting a wake phrase (e.g., “*Hi, Smart Glass*”), the system transitions into command mode. The user can then issue a spoken command, which is captured and sent to the base station for processing. At the base station, a **speech recognition model** interprets the user’s words, determines and distributes the requested task, and extracts the relevant keywords. In our design, there are two primary downstream tasks: **1) Obstacle-Aware Navigation**: mobility assistance and safe navigation. **2) Scene Description**: contextual awareness and object recognition. The natural language interface allows smooth task triggering: for instance, “*Navigate to the EECS building*” initiates navigation, while “*What’s in front of me?*” requests a scene description. Because both tasks involve computation-intensive AI processing, they are designed to run mutually exclusively to reduce base-station cost and maintain system responsiveness.

1. *Obstacle-aware Navigation*

The obstacle-aware navigation function is designed to ensure safe and efficient mobility by combining GPS-based route guidance with real-time obstacle avoidance. GPS navigation, implemented on the base station, provides high-level wayfinding similar to conventional map applications, announcing directions through audio prompts such as “*Turn left in ten meters.*” or “*Walk straight for five meters.*” Obstacle avoidance addresses immediate safety by detecting hazards in the user’s path and providing timely alerts. Instead of relying on computationally expensive large models, the system adopts a lightweight solution using pre-trained models such as [Ultralytics YOLO](#), trained on the [COCO dataset](#). This enables real-time detection of common obstacles like cars, bicycles, and pedestrians, while maintaining computational efficiency. Once detected, objects are analyzed to estimate their distance and orientation, allowing the system to determine whether intervention is required.

To convey information to the user, the system integrates two complementary feedback channels: audio feedback for GPS-based navigation instructions and tactile feedback for obstacle avoidance. Specifically, two micro-vibromotors are installed on the arms of the glasses to deliver intuitive haptic alerts, where vibration on the left arm indicates an obstacle to the left, vibration on the right indicates an obstacle to the right, and intensity reflects urgency, with stronger feedback signaling closer hazards. This design ensures that high-level navigation remains clear through audio, while obstacle alerts are communicated through low-latency, intuitive tactile cues that prioritize user safety.

2. *Scene Description*

The scene description function is designed to give users a concise and meaningful understanding of their surroundings, effectively mimicking the visual awareness normally available through human vision. This feature is powered by a lightweight vision–language model (VLM) that runs locally on the base station, taking both the camera image and a user prompt (e.g., “*Is there an empty chair?*”), as input and producing a short natural language description as output. The generated text is then converted to speech via a text-to-speech (TTS) module and streamed back to the smart glasses for real-time delivery. While compact VLMs necessarily sacrifice some depth and richness of description, models such as [LLaVA 1.6](#) have proven effective for simple, everyday scenes and can be deployed efficiently on embedded systems. Rather than aiming for exhaustive scene analysis, the goal is to deliver high-value, contextually relevant information that improves situational awareness, such as indicating the presence of a chair in a waiting room. By focusing on short, targeted descriptions, this functionality complements obstacle-aware navigation: navigation ensures safety and mobility, while scene description enhances general awareness and decision-making in daily activities. For demonstration purposes, we will showcase scenarios in which brief, well-timed descriptions add practical value, such as identifying items in a store, describing landmarks, or confirming the presence of objects in a room, highlighting the usefulness of this feature even with the constraints of lightweight VLMs.

B. The Smart Glass system (Customized PCB based on ESP32)

The smart glass system is built on a customized PCB shaped as a wearable frame, the places for glass lenses are filled with PCB so that there is enough room to place all the components. The system is powered by a USB Type-C cable connected to the base station, which is also used for data transmission. This system mainly does the following things:

- The processor continuously polls the wake word.

- Record the user's question in audio format and transmit it to the base station.
- Receive AI response in audio format from the base station.
- Play the response with a speaker.
- Take videos or pictures using the camera when necessary:
 - In obstacle-aware navigation mode, it continuously captures video for obstacle detection.
 - In scene description mode, it takes one picture for AI description.
- Transmit pictures to the base station.
- Receive obstacle avoidance alert from the base station.

1. Hardware Selection

Figure 3 shows the block diagram of the smart glass system, and the following sections provide specific details about the hardware selections.

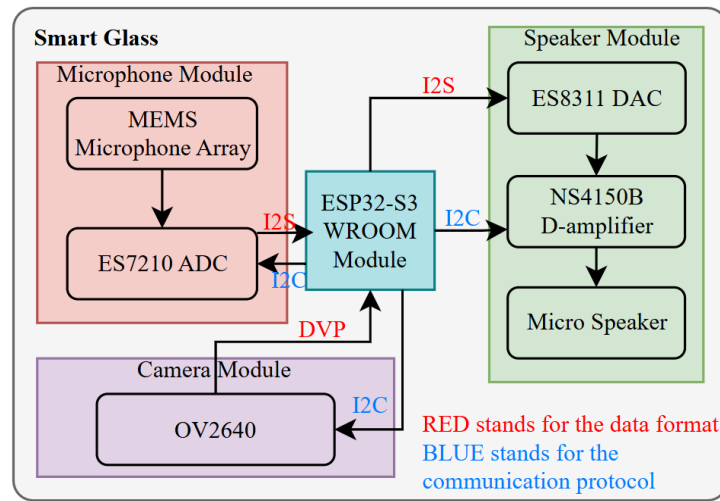


Figure 3: Functional block diagram of the smart glass

MCU targeted at Wakeword Recognition

We selected the **ESP32-S3 WROOM module combined with the WakeNet engine** for our system. It is compact and lightweight while still powerful enough to run wake word detection without the need for an external processor. Also, Espressif offers several development boards based on the ESP32-S3, which allow us to quickly prototype and validate functions. The ESP32-S3 ecosystem is mature and well supported. WakeNet, as an officially maintained wake word engine, has extensive documentation, community experience, and proven reliability. This strong technical foundation reduces development risk and ensures future scalability. Together, these advantages make the ESP32-S3 with WakeNet the most suitable choice for our project.

Microphone module

We selected a MEMS microphone array with the ES7210 audio ADC for our system. This combination provides stable and high-quality audio, as it integrates a low-noise preamplifier and high-performance ADC, ensuring consistent signal quality and a better signal-to-noise ratio, which is essential for accurate wake word detection. The ES7210 also enables acoustic echo cancellation (AEC),

beamforming, and noise suppression, allowing reliable voice detection even when the system plays audio. The ES7210 outputs ready-to-use I²S PCM data, offloading decoding from the ESP32-S3. With a MEMS microphone array, the system can suppress background noise and achieve far-field voice pickup up to 5 meters.

Camera module

We selected the OV2640 camera module for our system. This widely supported module has a low cost and low power consumption. It supports high resolution at 1600×1200, which is sufficient for face recognition, object detection, and basic navigation tasks. For obstacle-aware navigation, it can run smoothly at 640×480 or 800×600, providing enough clarity for YOLO-based models or similar detection frameworks. For scene description, its 1600×1200 still images offer adequate detail to support visual-language models. The module integrates a hardware JPEG encoder, which offloads image compression from the MCU. This reduces both processing and memory overhead on the ESP32-S3, making the system more efficient. Also, the OV2640 has extensive community resources, including open-source computer vision tasks.

Speaker module for audio feedback

We selected the CMS-2006-154SP micro speaker with the ES8311 DAC and the NS4150B Class-D amplifier. This design is compact, with only around 2 mm thick and 20 mm in diameter, making it easy to integrate into the glasses frame and PCB. The design has low power consumption, and a clear voice range between 500Hz and 4 kHz. The module is reliable and safe with built-in protection (overcurrent, overheating, undervoltage) for stable operation.

Vibromotor for tactile feedback

We selected a coin-type vibration motor for tactile feedback. Its flat and compact design makes it easy to embed in the glasses frame, while providing clear but subtle vibrations that effectively guide visually impaired users. It also features low power consumption and proven reliability, making it ideal for continuous wearable use.

2. *Video Frame Rate*

A video stream is required for real-time obstacle avoidance. Both the memory of the MCU and communication bandwidth of the USB interface place constraints on the video frame rate that can be achieved. The ESP32-S3 microcontroller used in the smart glass system is equipped with 16 MB of external PSRAM, which provides sufficient buffer space for multimedia tasks despite the chip's limited on-chip memory. In the most demanding obstacle-aware navigation mode, the OV2640 camera streams compressed JPEG frames while audio feedback is simultaneously received from the base station. Given the ESP32-S3's USB Full-Speed interface (~1 MB/s effective throughput), practical frame rates are constrained by link bandwidth rather than memory: VGA (640×480) frames compress to ~30–50 kB, allowing sustained transmission at 25~30 fps, while higher resolutions such as SVGA or XGA remain feasible at 15~20 fps. Audio adds only a negligible overhead (<5% of bandwidth). With careful memory allocation and frame-by-frame streaming, the system can reliably support real-time video and audio for robust navigation assistance.

3. PCB Design

Finally, transitioning from development boards to a customized PCB requires careful hardware integration. The bare ESP32-S3 chip must be mounted and connected in a way that preserves all critical functionality, including programmability and debugging capability, after assembly. Special attention will be paid to providing proper power regulation, USB connectivity, and peripheral interfaces, while adhering to specific design guidelines published by Espressif. Fortunately, comprehensive reference designs and PCB layout [guides](#) are available from the manufacturer, which will significantly reduce the risk of integration errors. With careful schematic capture and PCB layout, migration from the development board to a production-ready customized PCB should be straightforward.

4. Safety and Comfort Considerations

We will design a protective case, likely 3D-printed using TPU, that is soft and flexible enough to conform comfortably to the user's face. This enclosure will house the custom PCB securely inside while providing a compliant interface with the skin. The soft material not only ensures long-term comfort during wear but also serves as a safety measure, preventing the PCB from causing injury to the user in the unlikely event of mechanical failure or accidental impact.

C. The Base Station System (Jetson Nano based computing hub)

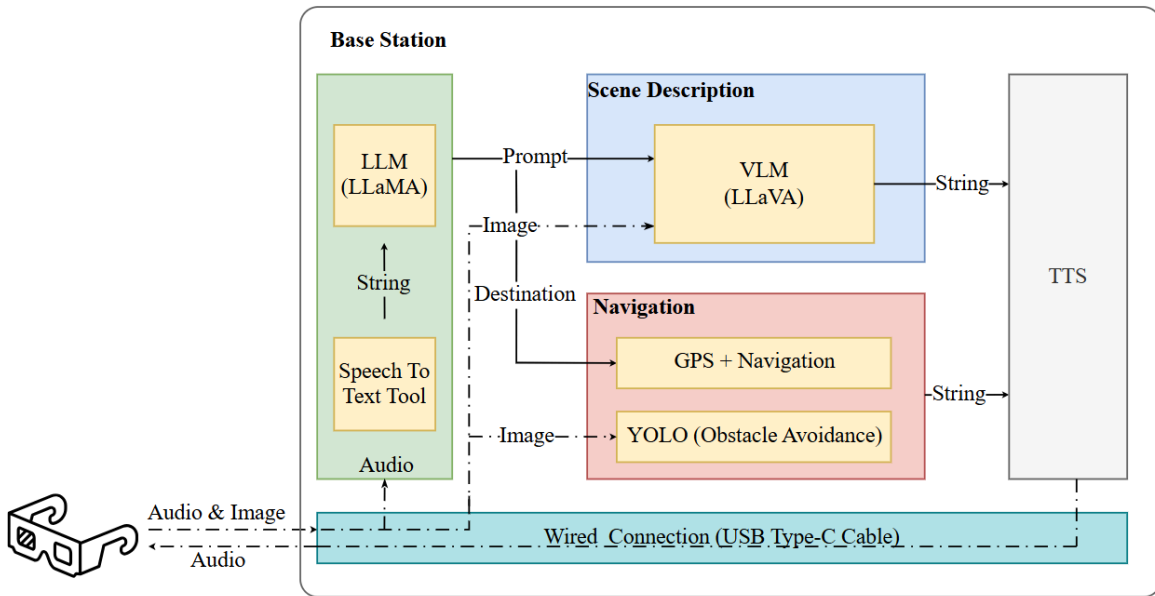


Figure 4: Functional block diagram of the base station

A key design decision in our system is the inclusion of a dedicated portable base station to handle AI computation. While commercial embedded microcontrollers are efficient for low-power tasks such as wake word detection, they lack the processing capability required for real-time natural language understanding, obstacle detection, and scene analysis. An alternative would be to offload computation to the cloud; however, this approach presents several limitations. First, transmitting continuous audio and video to external servers raises significant privacy and security concerns, as sensitive personal data could be exposed. Second, reliance on cloud services introduces latency that can degrade responsiveness, which is unacceptable in safety-critical tasks such as obstacle avoidance. Finally, cloud-based solutions depend

on stable internet access, which may not be available in all environments, especially outdoors or in transit. By contrast, a local embedded platform provides a secure, low-latency, and always-available solution, ensuring the system can function reliably in real-world conditions.

The base station system is an off-the-shelf development board – Nvidia Jetson Orin Nano developer kit. It is a powerful embedded computer where we can deploy AI applications like YOLO and LLaVa locally. Figure 4 shows the function block diagram of the base station, and the following sections describe each block in detail, illustrating how the base station is able to support all the computation needed for the smart glasses. The final section analyzes the workload to get all these modules function properly, and verify that we can achieve some meaningful goals only by using open source AI modules that have already been trained and validated.

1. Speech Recognition

The speech recognition module is responsible for processing the user’s spoken input after the wake word has been detected. Implementation will rely on lightweight speech-to-text models optimized for embedded platforms. One potential solution is to use [Whisper small](#) or similar models fine-tuned for constrained devices, while integrating a lightweight LLM to produce structured outputs that can be automatically parsed by downstream modules. The Jetson Orin Nano has demonstrated the ability to run compact LLMs such as LLaMA 3.0 with reasonable response times, making it a feasible platform for this task. To ensure safety and reliability, a text filtering layer will be added to validate and sanitize model outputs before they are passed downstream, reducing the risk of faulty behaviors or misinterpretations.

2. Obstacle Detection Module for obstacle avoidance

The obstacle detection module provides real-time awareness of hazards in the user’s immediate path. Implementation is based on pre-trained object detection models such as YOLO11x, which have been trained on the COCO dataset. The advantage of YOLO is its balance between accuracy and real-time performance, making it suitable for embedded deployment. The Jetson Orin Nano is capable of running optimized versions of YOLO at acceptable frame rates, enabling real-time obstacle detection. The system can also dynamically adjust frame resolution and processing frequency to maintain responsiveness while conserving power.

3. Vision Language Model for Scene Description

The VLM enables non-navigational assistance by generating textual descriptions of the environment based on both images and user prompts. Implementation relies on lightweight VLMs such as LLaVA 1.6, which are compact enough to run locally on the Jetson Orin Nano. These models accept an image and a textual query as input and produce a short descriptive text string as output. Although smaller than state-of-the-art multimodal models, they provide sufficient functionality for everyday use cases such as recognizing common objects, identifying signs, or describing the contents of a storefront. The generated descriptions are then passed to a text-to-speech system to complete the interaction loop.

4. GPS locomotion and navigation

The GPS module provides high-level wayfinding capabilities by tracking the user’s location and guiding them toward their specified destination. Implementation involves a GPS module like [NEO-6M](#), directly connected to the base station to provide location information (longitude and latitude) of the user.

Then, an open source navigation software like [navit](#) will be run on the base station to provide directional instructions.

5. Text-to-Speech module

The text-to-speech module is basically the last gate before any message is transmitted to the smart glass system. Compact, open-source solutions such as Coqui TTS or ESPnet-based TTS can be adapted to run on the Jetson Orin Nano with reasonable performance. The TTS engine must support multiple voices and possibly adjustable speaking rates to accommodate user preferences. The audio stream is transmitted back to the glasses' speaker via USB, completing the interaction loop.

6. SRAM Limitation

We are aware of the 8GB memory limitations of the Jetson Orin Nano board. Given this, we cannot run very complex AI models locally. Besides, multiple models (speech recognition, obstacle detection, and vision language model) cannot be run simultaneously. A sequential execution strategy ensures that memory usage remains well below capacity at each stage, while still supporting all three core assistive functions, voice interaction, obstacle-aware navigation, and scene description, on-device without requiring Internet connectivity.

7. Power Management

The entire system will be battery-powered. The Jetson Orin Nano developer kit is the dominant load, consuming approximately 17–27W depending on operating mode, while the smart glasses—including the ESP32-S3, OV2640 camera, audio codec, speaker driver, and haptic feedback—consume less than 1W on average. GPS and other peripherals add only a few hundred milliwatts. In total, the system draws about 18W in standard operation and up to 28W under maximum performance. A high-capacity USB-C PD battery, such as the [Anker 737 Power Bank](#) (model A1289, 24,000mAh, 88Wh), can reliably supply 20V to the Jetson via a PD-to-barrel adapter and simultaneously power the glasses over USB. This configuration supports continuous operation for approximately 4.5 hours in normal mode and 3 hours in high-performance mode, providing sufficient runtime for field demonstrations and testing.

8. Workload Analyze

At first glance, to make both obstacle-aware navigation and scene description working properly seems daunting, given all the AI modules involved. Fortunately, after doing some research online, we found that most of the modules are available, and can be cloned from a github repository directly. The following is a list of all the applications we need to integrate on the base station:

- Text to speech model: <https://github.com/OHF-Voice/piper1-gpl>
- Speech to text model: <https://github.com/openai/whisper>
- Obstacle detection model: <https://github.com/ultralytics/ultralytics>
- Vision language model: <https://github.com/haotian-liu/LLaVA>
- GPS navigation software: <https://github.com/navit-gps/navit>

Other things that still need us to complete include: 1). An interface with the GPS module. 2). Communication interface with the ESP32. 3). A script integrating all the existing models.

IV. Way Forward and Schedule

A. Gantt Chart



B. Milestone Deliverables

1. Milestone 1 (Oct 9):

Major Deliverables:

- **The wake word detection module should be able to run on the ESP32**
Demonstration: An LED will be lit up when the key word is detected by the ESP32. The demo will be on the [ESP32-S3-KORVO-2](#) dev board.
- **Communication between ESP32 and Jetson Nano base station set up**
Demonstration: Audio & video can be transmitted from the ESP32 dev board to the Jetson Nano base station through USB. Audio can be transmitted from Jetson Nano to ESP32.

Minor Deliverables:

- VLM for scene description should be able to run on base station
Criteria: The VLM has been deployed to the base station and can respond to common image + text prompt input.
- Speech-to-Text module should be able to run on the base station
Criteria: The speech-to-text module has been deployed on the base station and can respond to common audio input.
- Text-to-Speech module should be able to run on the base station
Criteria: The text-to-speech module has been deployed on the base station and can generate audio output for any text input.
- Lightweight LLM should be able to run on the base station
Criteria: The LLM has been deployed to the base station and can respond to common text input.

2. Milestone 2 (Nov 4):

Major Deliverables:

- **GPS-based navigation module can work properly.**
Demonstration: The GPS module can send location (latitude and longitude) to the base station. A navigation software can produce text instructions based on GPS coordinates.
- **The obstacle detection module can work properly.**
Demonstration: The YOLO model can be successfully run on the base station, and take video input from the ESP32, and label the obstacles (bicycles, car, person) detected. The demo will be done with the ESP32 dev board.

Minor Deliverables:

- PCB has been designed and the order has been placed
Criteria: The microphone module, camera module and speaker module has been integrated with ESP32-S3 WROOM module. The schematic drawing and routing have been finished and the order has been placed.
- Scene description module can work properly
Criteria: Audio recorded from the ESP32 can be sent to the VLM model deployed on the base station. A text output describing the image related to the audio prompt can be generated by the VLM.
- Design of eyeglass frame has been finished

Criteria:

- Check in on a wooden/plastic/3D printed frame of the glasses with place to put glasses PCB;
- Check in on a wooden/plastic/3D printed frame of the base station with a place to put the processor and power supply safely.
- An LED can light up in coordination with the camera
Criteria: when the camera is working, an LED can be lit up.
- IMU module implemented
Criteria: IMU module can sound the speaker if significant impact, which indicates the glasses are fallen, is detected.

V. Design Expo and Final Presentation

What We'll Highlight in the Video & Demo Flow

We'll build a **multi-view video montage** plus a **live walk-through** to showcase system capabilities in a polished, immersive way. Here's how to stage it:

Functions	Videos that play in a loop	Live Demo	Audience Takeaway
Scene Description	We will show an application scenario for assisting the visually impaired in finding seats by playing two combined views: 1. Footage from third-person view in the office 2. camera feed overlay (glass POV) + show model output (bounding boxes + description text) → The wearer says "Is there an empty chair" and the glass replied "Yes, there is an empty chair by the back wall"	We will show how the core model (VLM) works here : Give the prompt as "Describe the image concisely" and show different objects in front of the camera → Show the VLM's text output on the display ("A man is holding a black mug") You can get an intuition from this video	Demonstrates the VLM / vision → spoken feedback chain; helps visitors "see through your eyes"
Navigation + Obstacle Avoidance	The wearer walks through a clear floor with 3-4 soft foam or cone obstacles, maybe a "person" (cardboard stand) placed in the path. At the same time display three views: 1. Third-person "chase" view as the wearer walks 2. Glasses POV (camera) with overlay: YOLO bounding boxes and vibro status text ("Left motor = ON", "Right motor = OFF") 3. Wireframe or split showing	The wearer walks slowly through the same setup guided by motor cues and audio prompts: the system should detect obstacles, vibrate the correct side, say "Obstacle ahead, step right" or "Obstacle cleared.". The three views are overlaid again on an external display. Use a spotter for safety but keep minimal interference	Shows live mapping from vision → control → haptics + audio; supports the narrative of obstacle awareness in real space

	basestation YOLO output and navigation decisions (e.g. arrows, path) Overlay the audio cues spoken & vibromotor beep them		
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VI. Budget and Material

A. BOM Table

#	Sub-assembly / Part	Qty	Unit \$	Ext. \$	Likely Vendor	Note / Justification
1	NVIDIA Jetson Orin Nano Super Dev Kit	1	249	249	NVIDIA	This is the unit you call out for on-device YOLO/VLM
2	ESP32-S3-KORVO-2 Dev Board	1	46.3	46.3	Mouser	Use as test board for your custom PCB
3	ESP32-S3-WROOM-1-N16 module	2	34.8	79.6	Amazon	For the glasses PCB
4	ES7210 4-ch Audio ADC (QFN-32)	2	0.42	0.84	LCSC	Wake-word mic array
5	INMP441 I ² S MEMS mic modules	1	10	10	Amazon	For bench test
6	CMS-2006-154SP 20 mm speaker	2	2.87	5.74	Digi-Key	Audio out
7	NS4150B 3 W Class-D amp (MSOP-8)	4	0.06	0.24	LCSC	Amplifier
8	OV2640 camera module (DVP/mini module)	2	8	16	Arducam via Robotshop	Vision
9	Coin vibration motor, 10 mm, 3 V (e.g., Vybionics VC1034B018F)	4	2	8	Digi-Key	Haptics
10	u-blox NEO-6M (or M8N) GPS module	1	19	19	Amazon	Positioning (base station)
11	USB-C to USB-C data cable, 10 Gbps (1 m)	2	13	26	Amazon	Power / data cable
12	65 W USB-C PD Power Bank (20,000 mAh)	1	47	47	Amazon	Power supply

13	Breadboard + jumper kit	1	10	10	Amazon	Product prototype / test
14	Resistor assortment	1	7	7	Amazon	Prototype / test
15	LED assortment	1	7	7	Amazon	Prototype / test
16	Prototype PCB fab	1	50	50	JLC	PCB fabrication
17	Misc.	1	60	60	Amazon / LCSC	Miscellaneous
18	Total	/	/	641.72	/	/

B. Explanation of Part Selection

ESP32-S3 + WakeNet: fits always-on wake word at low power in the glasses; abundant examples + community.

ES7210 mic ADC: multi-mic array capture with good SNR and AEC/beamforming support; outputs I²S PCM, offloading the S3.

OV2640: low-cost, integrated JPEG, adequate resolutions (UXGA stills / VGA-SVGA video) for detection and VLM snapshots.

CMS-2006-154SP + NS4150B: thin 20 mm speaker and efficient 3 W class-D amp for clear TTS prompts in a slim frame.

Coin vibromotors: reliable haptic channel with intuitive left/right cues for obstacle avoidance without cognitive load.

Jetson Orin Nano: local YOLO + compact VLM + TTS/ASR sequentially (memory-aware), addressing privacy/latency.

Please refer to the [Shopping List In Progress](#) document.

VII. Conclusion

Our project aims to deliver a practical, user-centered solution that enhances the independence and safety of people with visual impairments by combining navigation, obstacle avoidance, and scene description into a single assistive platform. At a high level, the system provides an intuitive, voice-driven interface that allows users to interact hands-free, receive real-time obstacle-aware navigation and scene description, all without relying on cloud connectivity. To achieve this, we integrated lightweight smart glasses detecting wake words, and a portable Jetson-based base station capable of running wake word detection, speech recognition, YOLO-based obstacle detection, and a compact vision–language model sequentially. The design carefully balances computational efficiency, latency, and power consumption, with all processing performed locally to ensure privacy and reliability. Challenges such as embedded memory constraints, maintaining low-latency video processing for obstacle detection, and achieving sufficient battery life are acknowledged but mitigated through hardware selection, optimized software pipelines, and a modular, scalable architecture.

By combining robust sensing, local AI computation, and multi-modal feedback, our system creates a compelling prototype that demonstrates how accessible technology can improve mobility and situational awareness for visually impaired users. This approach not only addresses an urgent real-world need but also lays a foundation for future enhancements, such as multi-task concurrency, richer scene descriptions, and improved miniaturization, making it a promising step toward more intelligent, practical, and inclusive assistive devices.

Appendix

A. Design & Engineering Criteria

Design Criteria	Importance	Will/Expect/Stretch
Voice Interaction between User and System should be functional	Fundamental	Will
Scene Description should be Authentic	Fundamental	Will
Power System should be functional	Fundamental	Will
Scene Description should include instructions to avoid obstacles	Important	Expect
An LED indicator should light up when camera is working	Optional; Legally important	Will
GPS based navigation should be functional in optimal satellite condition	Important	Expect
Voice activation module should be functional	Optional; can be replaced by button if unachievable	Expect
IMU should sound the speaker if glasses fell	Optional	Stretch
IMU should send an alert through SMS	Optional	Stretch

B. Milestone Deliverables

<u>Milestone 1:</u>
The wake word detection module should be able to run on the ESP32

Communication between ESP32 and Jetson Nano base station set up
VLM for scene description should be able to run on base station
Speech-to-Text module should be able to run on the base station
Text-to-Speech module should be able to run on the base station
Lightweight LLM should be able to run on the base station
<u>Milestone 2:</u>
GPS-based navigation module can work properly.
The obstacle detection module can work properly.
PCB has been designed and the order has been placed
Scene description module can work properly
Design of eyeglass frame has been finished
An LED can light up in coordination with the camera
IMU module implemented

C. Interface Design

[Base station interface](#)

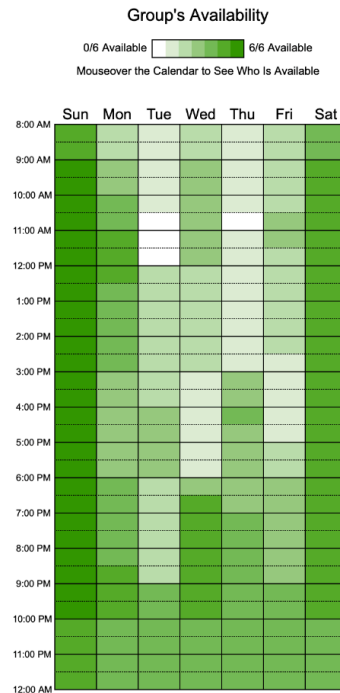
[Smart glasses interface](#)

D. Group Agreement

1. Workload Expectation

- Hours per person per week: ~20–24 hours (including meetings, individual tasks, and group work).
- Flexibility: Hours may increase slightly during debugging (early November) and before Expo (late November – early December).

2. Availability



3. Core Working Times

- Planned work blocks:
 - Group sessions: Sunday 10 A.M. to 6 P.M.
 - Individual work: Members continue tasks on their own time, but submit progress updates before each group session.
- Work cadence: Most hardware integration will be done during Sunday work sessions (access to lab/tools), while software development will progress individually during the week.
- Team Meeting: Sunday 10 A.M. to 6 P.M., same time as group sessions;
 - If there is need to hold extra meetings, discuss for a time that most people are available
- Known conflicts:
 - Ruopu Dong: Family Gathering on Nov. 29
 - Jinlin Li: TOEFL test on Oct. 30
 - Haobo Fang: GRE test on Nov. 1

4. Group vs. Individual Work

- Group Work (collaborative):
 - **Build Program Hierarchy**
 - **Debugging**
- Subgroup - embedded - work:
 - Glasses prototyping
 - Glasses PCB design
 - Base station prototyping
 - Glasses - base station interconnection

- Subgroup - software - work:
 - Implement VLM
 - Implement speech-to-text model
 - Implement text-to-speech module
 - Implement lightweight LLM
 - Implement YOLO
 - Implement navigation software
 - Apply GPS module to support navigation software
- Individual Work (optional):
 - LED indicator
 - IMU impact detection

5. Roles & Responsibilities

- Embedded Subgroup: Ruopu Dong, Guanyu Xu, Zhuoyang Chen
- Software Subgroup: Haobo Fang, Yizhe Shen, Jinlin Li

Signatures:

*Jinlin Li, Yizhe Shen, Haobo Fang,
Zhuoyang Chen, Guanyu Xu, Ruopu Dong*